

GROUP WORK PROJECT # 1

MScFE 622: Stochastic Modeling

GROUP NUMBER: 16855

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Note: You may be required to provide proof of your outreach to non-contributing members upon request.

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Executive summary

This report presents the assumptions, results, interpretations, and figures for all project questions. SM Energy spot is \$232.90; the annual risk-free rate is 1.50%; one year is 250 trading days; and no dividend yield is assumed.

Step 1 - 15-day derivative

1(i) Heston calibration using Lewis (2001)

Heston represents equity returns with a randomly changing, mean-reverting variance process that can reproduce volatility clustering and implied-volatility skew. Lewis converts its characteristic function into a one-dimensional Fourier pricing integral.

Parameters (Equations 1-3). S_t is the stock price and v_t is its instantaneous variance at time t ; r is the continuously compounded risk-free rate and dt is an infinitesimal time increment. W_t^S and W_t^v are Brownian motions driving price and variance. The variance parameters are mean-reversion speed κ , long-run variance θ , volatility of variance σ , correlation ρ between the Brownian shocks, and initial variance v_0 .

Equation 1

$$dS_t = rS_t dt + \sqrt{v_t} S_t dW_t^S.$$

Equation 2

$$dv_t = \kappa(\theta - v_t) dt + \sigma\sqrt{v_t} dW_t^v.$$

Equation 3

$$dW_t^S dW_t^v = \rho dt.$$

Parameters (Equation 4). $\phi_T(z)$ is the characteristic function of the log return over maturity T at complex Fourier argument z ; $i=\sqrt{-1}$, S_0 and S_T are the initial and terminal stock prices, and E^Q denotes expectation under the risk-neutral measure Q .

Equation 4

$$\phi_T(z) = E^Q[e^{iz \log(S_T/S_0)}].$$

Parameters (Equations 5-6). C_0 and P_0 are today's European call and put values; S_0 is spot, K is strike, T is time to maturity, and r is the risk-free rate. u is the Fourier integration variable, $i=\sqrt{-1}$, $\text{Re}[\cdot]$ takes the real part, π is the circle constant, and ϕ_T is the Heston characteristic function from Equation (4).

Equation 5

$$C_0 = S_0 - \frac{e^{-rT} \sqrt{S_0 K}}{\pi} \int_0^\infty \text{Re} \left[\frac{e^{iu \log(S_0/K)} \phi_T(u - i/2)}{u^2 + 1/4} \right] du.$$

Equation 6

$$P_0 = C_0 - S_0 + Ke^{-rT}.$$

Parameters (Equation 7). Θ is the trial Heston parameter vector; $N=10$ is the number of option quotes and i indexes them. $V_i^{\text{model}}(\Theta)$ and V_i^{market} are the model and observed USD prices, respectively, and MSE is their mean squared pricing error.

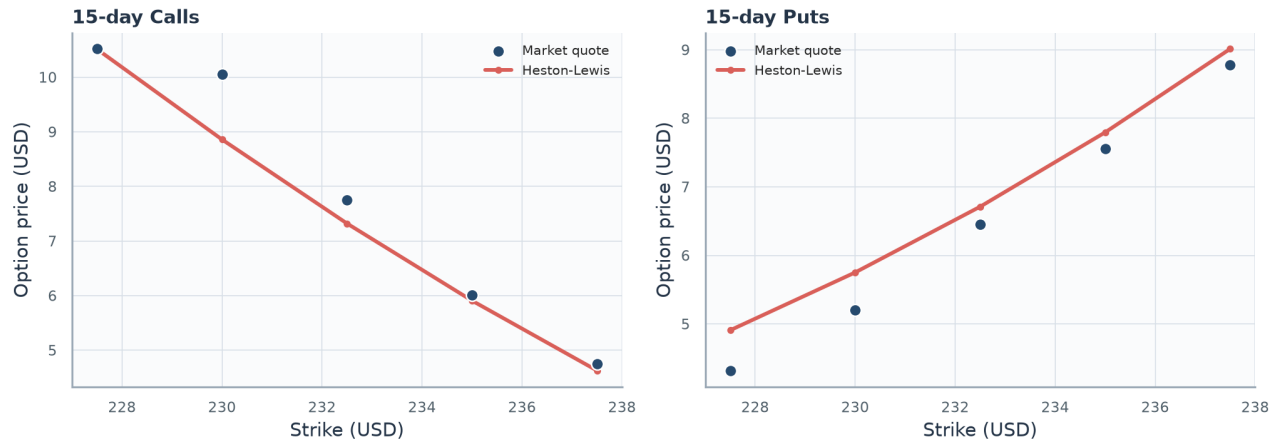
Equation 7

$$\text{MSE}(\Theta) = \frac{1}{N} \sum_{i=1}^N (V_i^{\text{model}}(\Theta) - V_i^{\text{market}})^2, \quad N = 10.$$

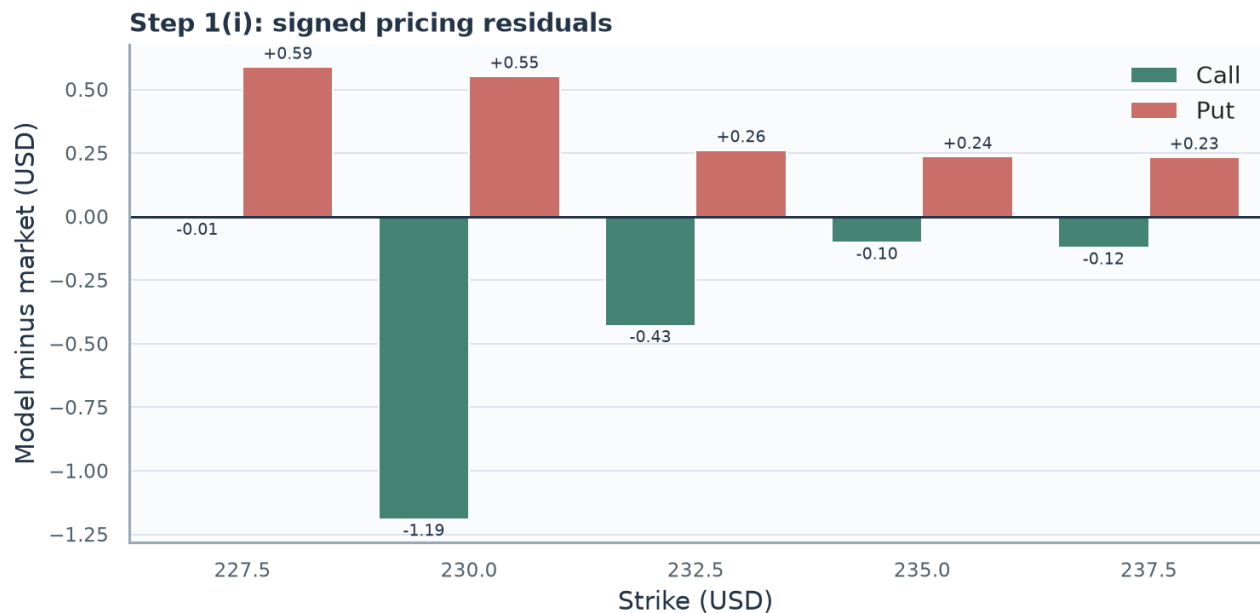
The 15-day calls and puts were fitted jointly using the Heston specification and Lewis representation (Heston 327-343; Lewis), with unweighted USD price MSE as required by the project brief (WorldQuant University 1-2). The fitted parameters are $\kappa=2.2032$, $\theta=0.0010$, $\sigma=1.2733$, $\rho=-0.9444$, and initial variance=0.1084. MSE is 0.2458 and RMSE is \$0.4958.

The input quotes have a put-call parity RMSE of \$0.9068; an arbitrage-consistent fit cannot make all residuals zero. Short maturity also weakens identification of long-run variance and mean reversion.

Initial variance implies 32.93% annualized volatility and $\rho = -0.9444$ indicates a strong leverage effect. Theta is at its lower bound and the Feller diagnostic is -1.6170; the parameter vector should therefore be treated as a short-horizon fit rather than a long-run forecast.



Plot 1. 15-day market and model prices



Plot 2. 15-day residuals

1(ii) Carr-Madan comparison

Carr-Madan damps the option-price function so it can be evaluated through an integrable Fourier transform, providing an independent pricing and calibration route for the same Heston dynamics.

Parameters (Equations 8-9). $\psi_T(v)$ is the damped transform at Fourier frequency v and maturity T ; $\alpha > 0$ is the damping parameter, r is the risk-free rate, and $i = \sqrt{-1}$. $\phi_{\log S_T}$ is the characteristic function of the terminal log price. $C_0(K, T)$ is today's call value, K is strike, and $\text{Re}[\cdot]$ takes the real part of the integrand.

Equation 8

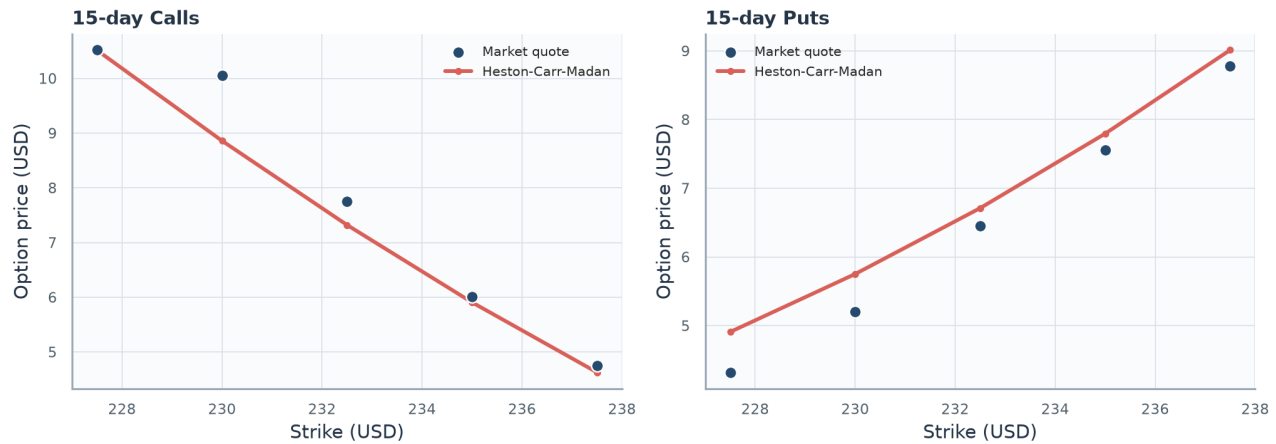
$$\psi_T(v) = \frac{e^{-rT} \phi_{\log S_T}(v - (\alpha + 1)i)}{\alpha^2 + \alpha - v^2 + i(2\alpha + 1)v}.$$

Equation 9

$$C_0(K, T) = \frac{e^{-a \log K}}{\pi} \int_0^\infty \text{Re}[e^{-iv \log K} \psi_T(v)] dv.$$

The damped transform follows Carr and Madan (61-73). Its MSE is 0.2458, compared with 0.2458 for Lewis. Fitted prices and numerical stability are the meaningful comparison.

Kappa changes from 2.2032 under Lewis to 0.0504 under Carr-Madan even though the losses are identical to four decimals. This is consistent with weak short-maturity identification: the two numerical formulas agree on prices, while the optimizer can move along a flat parameter direction.



Plot 3. 15-day Carr-Madan fit

1(iii) 20-day ATM Asian call

The arithmetic-average Asian payoff reduces dependence on a single terminal stock price. Monte Carlo simulation is required because the payoff depends on the complete daily path.

Parameters (Equations 10-13). A_T is the arithmetic average through maturity T ; $n=20$ is the number of future daily observations, j indexes observation dates t_j , and S_{t_j} is the stock price at each date, including $S_{t_0}=S_0$. Π_T is the Asian-call payoff and $K=S_0$ is its at-the-money strike. $\widehat{V}_0$ is the discounted Monte Carlo value, $M=200,000$ is the number of paths, m indexes paths, $\Pi_T^{(m)}$ is path m 's payoff, and r is the risk-free rate. V_{client} is the quoted value after multiplying fair value by 1.04 to include the 4% fee.

Equation 10

$$A_T = \frac{1}{n+1} \sum_{j=0}^n S_{t_j}, \quad S_{t_0} = S_0.$$

Equation 11

$$\Pi_T = \max(A_T - K, 0), \quad K = S_0.$$

Equation 12

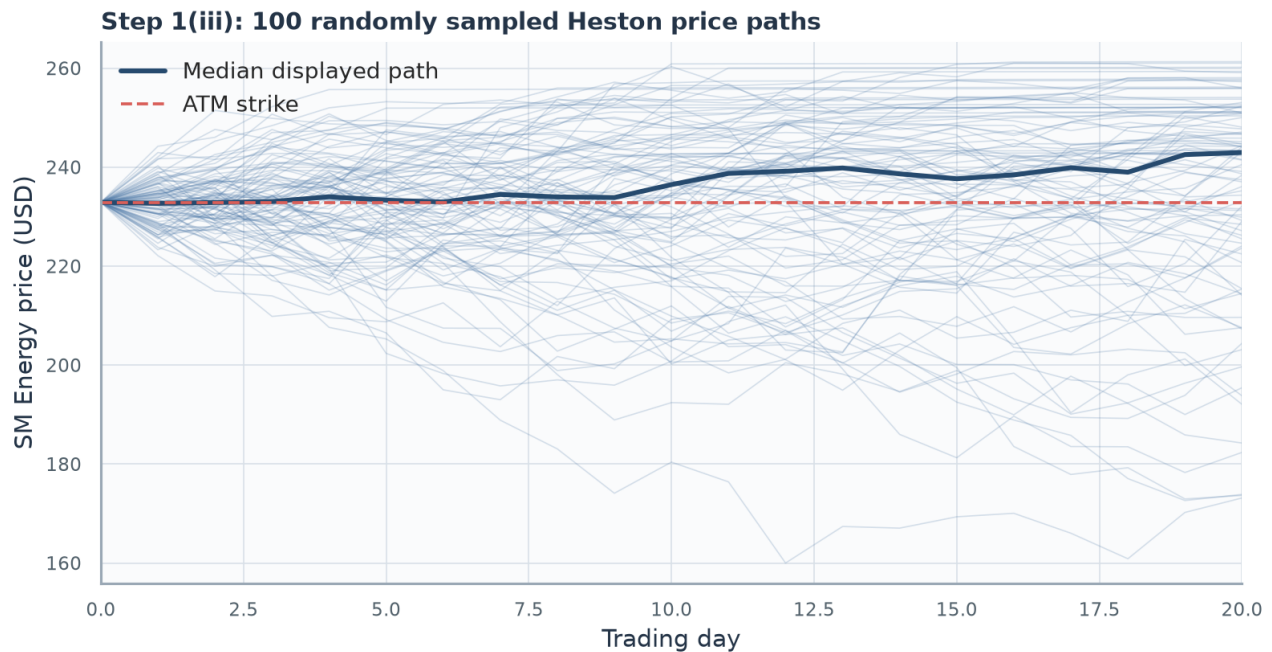
$$\widehat{V}_0 = e^{-rT} \frac{1}{M} \sum_{m=1}^M \Pi_T^{(m)}.$$

Equation 13

$$V_{\text{client}} = 1.04 \widehat{V}_0.$$

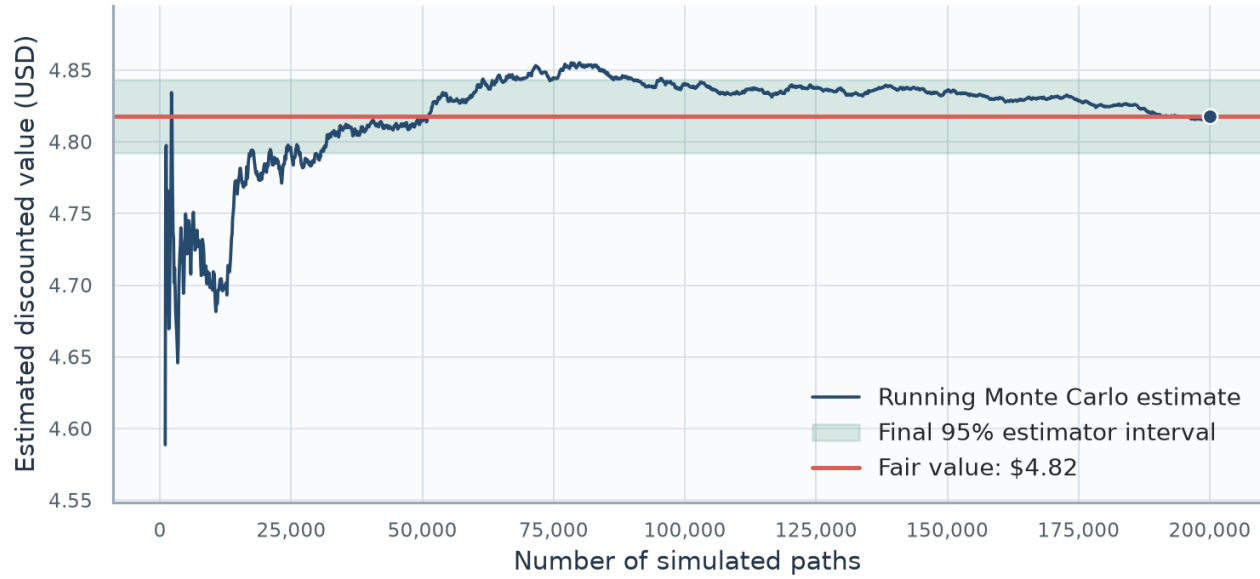
Today's spot is included in the arithmetic average, following the project instructions (WorldQuant University 1-2). Across 200,000 risk-neutral scenarios, fair value is \$4.8177 (standard error \$0.0130), with a 95% estimator interval of \$4.7922 to \$4.8432. The 4% fee produces a client price of \$5.0104.

The standard error is 0.27% of fair value, so simulation noise is small relative to calibration risk. The interval measures estimator precision and is not a guaranteed payoff range for the client.



Plot 4. 100 randomly sampled Asian-call paths

Step 1(iii): Monte Carlo convergence to Asian-call fair value



Plot 5. Asian-call fair-value convergence

Step 2 - 60-day derivative

2(i) Bates calibration using Lewis

Bates extends Heston with random discontinuous price jumps, allowing the model to represent rare abrupt moves in addition to ordinary stochastic-volatility dynamics.

Parameters (Equations 14-16). S_{t^-} is the stock price immediately before a jump; r , v_t , and W_t^S retain their Heston meanings. N_t is a Poisson jump count with intensity λ , $J=e^Y$ is the multiplicative jump size, and Y is normal with mean μ_J and standard deviation δ_J . The compensator $\kappa_J=E[J-1]$ preserves the risk-neutral drift. $\phi_J(u)$ is the jump characteristic-function multiplier at Fourier argument u over maturity T , with $i=\sqrt{-1}$.

Equation 14

$$\frac{dS_t}{S_{t^-}} = (r - \lambda \kappa_J) dt + \sqrt{v_t} dW_t^S + (J - 1) dN_t.$$

Equation 15

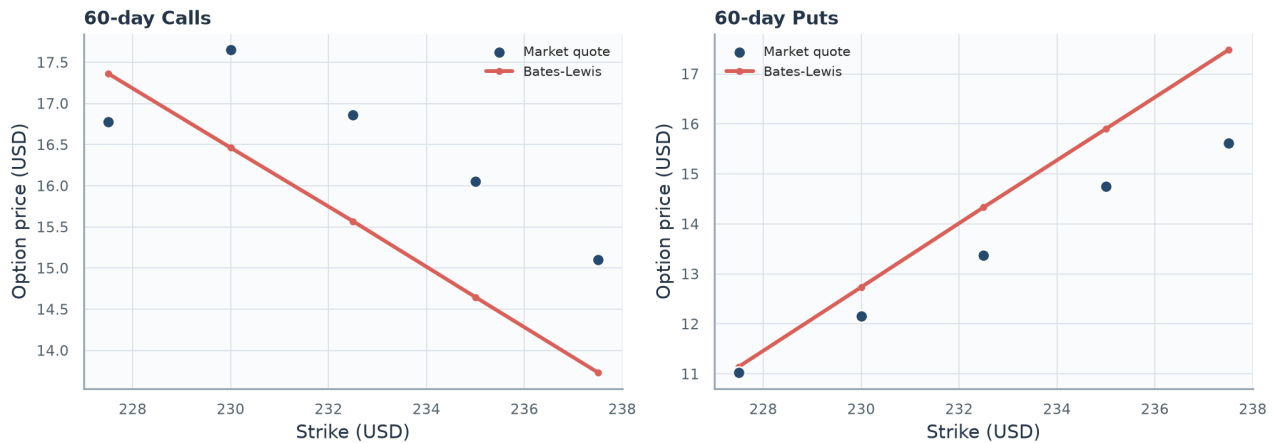
$$\kappa_J = E[J - 1] = \exp\left(\mu_J + \frac{\delta_J^2}{2}\right) - 1.$$

Equation 16

$$\phi_J(u) = \exp\left\{\lambda T \left[\exp\left(iu\mu_J - \frac{1}{2}\delta_J^2 u^2\right) - 1 - iu\kappa_J\right]\right\}.$$

The 60-day jump-diffusion calibration follows Bates (69-107) and has MSE 1.3324 and RMSE \$1.1543.

The call-put pairs have parity RMSE \$2.2533. Rho is at its upper bound, theta and initial variance are at lower bounds, and jump volatility is at its lower bound. These boundary estimates and a Feller diagnostic of -0.0228 show that the eight parameters are weakly identified by this small, internally inconsistent cross-section.



Plot 6. 60-day market and model prices

2(ii) Carr-Madan comparison

This specification retains the Bates jump and variance dynamics but uses the Carr-Madan transform as a numerical cross-check of Lewis pricing.

Parameters (Equations 8-9). $\psi_T(v)$ is the damped transform at Fourier frequency v and maturity T ; $\alpha > 0$ is the damping parameter, r is the risk-free rate, and $i = \sqrt{-1}$. $\phi_{\log S_T}$ is the characteristic function of the terminal log price. $C_0(K, T)$ is today's call value, K is strike, and $\text{Re}[\cdot]$ takes the real part of the integrand.

Equation 8

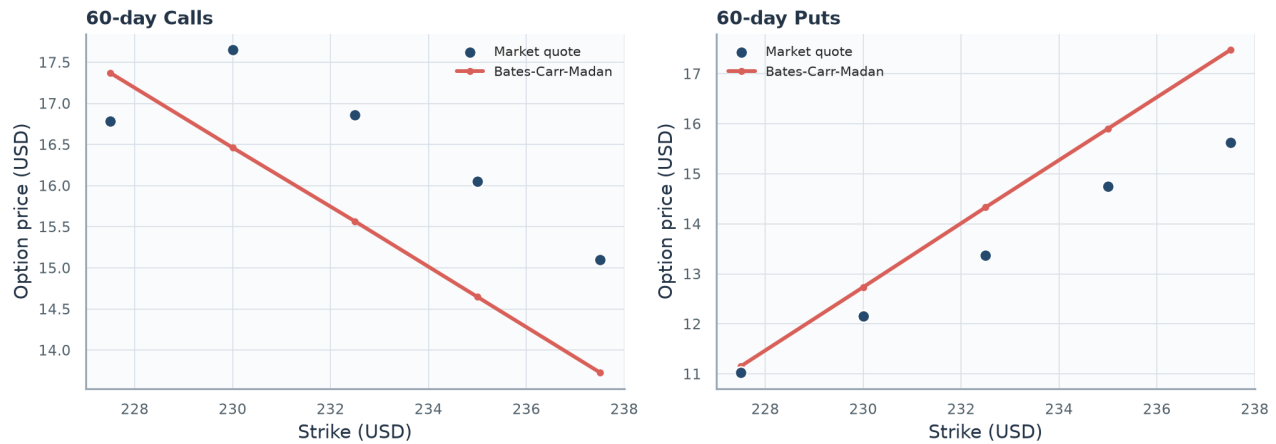
$$\psi_T(v) = \frac{e^{-rT} \phi_{\log S_T}(v - (\alpha + 1)i)}{\alpha^2 + \alpha - v^2 + i(2\alpha + 1)v}.$$

Equation 9

$$C_0(K, T) = \frac{e^{-a \log K}}{\pi} \int_0^\infty \text{Re}[e^{-iv \log K} \psi_T(v)] dv.$$

The matching Carr-Madan calibration follows Carr and Madan (61-73) and has MSE 1.3335. The comparison is based on fitted-price errors and numerical stability.

Jump intensity and mean jump size remain stable across methods, but κ changes from 0.0114 to 14.9006. Nearly equal loss with very different mean reversion indicates a flat or multi-modal objective; fitted prices are more reliable than parameter equality.



Plot 7. 60-day Carr-Madan fit

2(iii) 70-day 95%-moneyness put

The project specifies a strike of 95% of spot (WorldQuant University 3), giving \$221.25. Fair value is \$8.8187 and the price after the 4% fee is \$9.1715.

Parameters (Equation 17). K is the put strike, $S_0 = 232.90$ is the current SM Energy spot price, and 0.95 is the assignment's 95% moneyness factor, giving $K = 221.255$.

Equation 17

$$K = 0.95S_0 = 0.95(232.90) = 221.255.$$

The 70-day maturity is a ten-trading-day extrapolation beyond the calibration sample. The extension is modest, but the boundary-sensitive Bates estimates make calibration risk more material than the difference between the two Fourier integration methods.

Step 3 - Euribor rates

3(i) CIR calibration

CIR models a non-negative, mean-reverting short rate and connects its parameters to the Euribor curve through closed-form zero-coupon bond prices.

Parameters (Equations 18-23). r_t is the short rate at time t , r_0 is its initial value, a is mean-reversion speed, b is the long-run rate, η is rate volatility, and W_t is a Brownian motion. $P(0,T)$ is today's zero-coupon bond price for maturity T ; $A(T)$ and $B(T)$ are the CIR affine bond coefficients and $\gamma = \sqrt{a^2 + 2\eta^2}$ is an auxiliary coefficient. $R(0,T)$ is the continuously compounded zero rate, while $\exp$ and $\log$ denote the exponential and natural logarithm.

Equation 18

$$dr_t = a(b - r_t) dt + \eta \sqrt{r_t} dW_t.$$

Equation 19

$$P(0, T) = A(T) \exp[-B(T)r_0].$$

Equation 20

$$B(T) = \frac{2(e^{\gamma T} - 1)}{(\gamma + a)(e^{\gamma T} - 1) + 2\gamma}.$$

Equation 21

$$A(T) = \left[\frac{2\gamma \exp((a + \gamma)T/2)}{(\gamma + a)(e^{\gamma T} - 1) + 2\gamma} \right]^{2ab/\eta^2}.$$

Equation 22

$$\gamma = \sqrt{a^2 + 2\eta^2}.$$

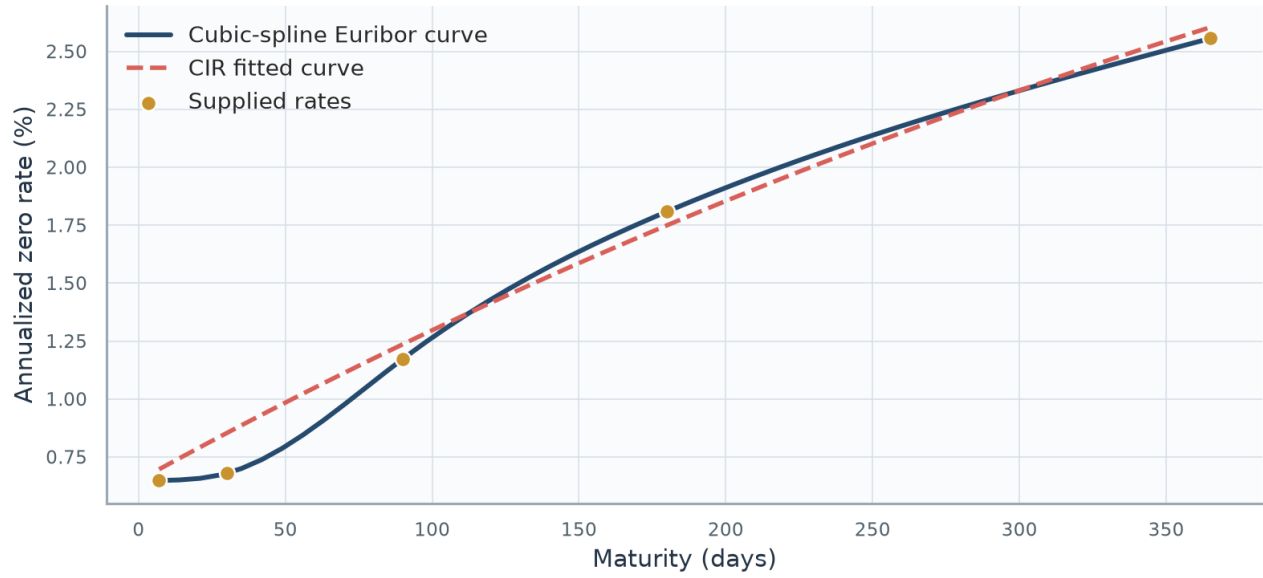
Equation 23

$$R(0, T) = -\frac{\log P(0, T)}{T}.$$

The supplied curve contains 0.648% (1 week), 0.679% (1 month), 1.173% (3 months), 1.809% (6 months), and 2.556% (12 months). Weekly interpolation follows the project brief (WorldQuant University 3-4), and the rate model follows Cox, Ingersoll, and Ross (385-407). The fitted values are $a=0.7525$, $b=0.0742$, $\eta=0.5638$; MSE is 0.00000062.

Mean reversion implies a half-life of 0.92 years, while the long-run rate 7.4170% lies above current 12-month Euribor. Curve RMSE is 7.90 basis points, but the Feller diagnostic -0.2063 is negative, so the process can reach zero and the broad dynamics remain uncertain.

Step 3(i): Euribor term structure and CIR fit



Plot 8. Euribor curve and CIR fit

3(ii) One-year rate scenarios

Daily CIR simulation turns the calibrated rate dynamics into a one-year distribution, supporting an expected-rate estimate and an explicit measure of rate uncertainty.

Parameters (Equations 24-25). r_{1y} is the simulated terminal rate after one year and $Q_p(\cdot)$ is the p-quantile operator, so $Q_{0.025}$ and $Q_{0.975}$ form the central 95% range. estimated $E[r_{1y}]$ is the Monte Carlo estimate of the expected terminal rate; $M=100,000$ is the number of paths, m indexes paths, and $r_{1y}^{(m)}$ is path m 's terminal rate.

Equation 24

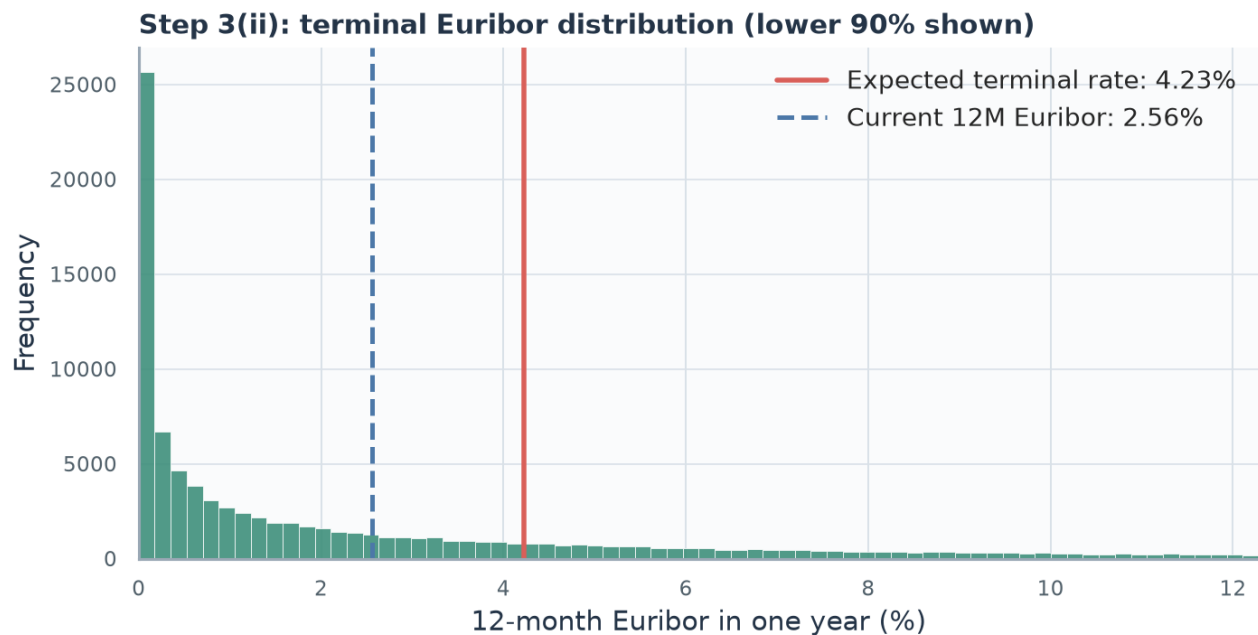
$$[Q_{0.025}(r_{1y}), Q_{0.975}(r_{1y})].$$

Equation 25

$$\hat{E}[r_{1y}] = \frac{1}{M} \sum_{m=1}^M r_{1y}^{(m)}, \quad M = 100,000.$$

Following the required simulation size (WorldQuant University 4), 100,000 daily scenarios give an expected terminal 12-month Euribor of 4.2289%; the 95% range is 0.0000% to 25.0856%. Higher expected rates increase discounting of future positive cash flows and generally lower present values, all else equal.

The expected rate is 167.3 basis points above today's 12-month quote. The distribution is strongly right-skewed: the lower endpoint is pinned at zero while the upper tail reaches high rates. Product effects are not universal because rates influence both discounting and risk-neutral drift; the CIR short rate is used here as a proxy for 12-month Euribor rather than a full multi-curve tenor model.



Plot 9. Zoomed terminal Euribor distribution

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